MILITARY SPECIFICATION

INDICATOR MONITOR TEST SET

ID 1956/ASW

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for the purpose of performing organizational maintenance of Link 4A Airborne Digital Data Communications equipment such as the AN/ASW-27B. The test set will decode the aircraft Reply Message data and display the data in numeric and binary forms. Control message data may also be received and displayed in a 42 bit binary display. The test set will also generate stored Control Messages from a read-only memory for transmission by the SM-51 1A/ASW. The stored messages provide the capability of transmitting a series of preprogrammed messages to fully exercise the Data Link equipment at organizational maintenance level.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Model Number	Nomenclature	Reference Paragraph
ID 1956/ASW	Indicator Monitor Test Set	3.7.1 through 3.7.1.12

1.3 Associated Equipment - The equipment shall operate with the following associated equipment shall operate with the following associated equipment:

(1) Digital Data Simulator (SM511A/ASW)
(2) Digits{ Data Recorder - Reproducer (RD-330/ASM).

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC-4935

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-T-5422 Testing, Environmental, Aircraft Electronic Equipment

MIL-E-7555 Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-T-8303 Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-T-21200 Test Equipment for Use with Electronic and Electrical Equipment, General Specification for

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-454 AN/ASW-27B Specification

MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment

MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

MIL-STD-794 Parts and Equipment, Procedures for Packing and Packaging of

MS17322 Meter, Time Totalizing, Miniature, Digital 115 Volt, 400 Cycle

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

STANDARDS

Naval Air Systems Command

AR-5 Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

AR-21 Support Equipment Design, Approval, Selection and Ordering for Bureau of Naval Weapons Contracts

WR-28 Message Standards for AN/USC-2, or Equivalent, Time Division Data Transmission System

2.1.1 Availability of Documents

When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Testing - This specification makes provision for testing (see 4).

3.2 Materials, Parts and Processes - The materials, parts, and processes shall be in accordance with MIL-T-21200.

3.2.1 Selection of Parts, Materials, and Processes - Except as specified herein, MI L-T-21200 shall apply for the selection of parts, materials and processes.

3.2.2 Approval of nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be in accordance with MIL-T-21200.

3.3 Design and Construction - The design and construction of the equipment shall be in accordance with MIL-T-21200.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 53 pounds.

3.3.2 Microelectronic Items - When used, microelectronic items shall be in accordance with AR-5.

3.3.3 Electron Devices - Electron tubes and semiconductor devices shall be in accordance with MIL-T-21200.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

3.3.4 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules. Modules shall normally be considered repairable.

3.3.5 Time Totalizing Meter- The equipment shall contain a time totalizing meter conforming to MS17322.

3.3.6 Cables and Connectors -The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.6.1 Cabling - The test equipment cabling shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested.

3.3.6.2 External Connectors - External connectors shall provide for the following:

Reference Designation	Connector Type	Function
J1	M39012/16-007	Receiver Data
J2	M23329/4-17	Stored Message Data
J3	MS3124E-16-26P	Data Link Data
J4	M23329/4-17	Record Data
J5	MS3124E-4P	Power

3.3.7 Controls - The control switches shall be in accordance with MIL-STD-454 Requirement 28, These controls shall be of such rugged design and construction that they will not be damaged when repeatedly operated under the specified service conditions.

3.3.8 Human Engineering - Human Engineering shall be in accordance with MIL-T-21200.

3.4 Reliability and Maintainability - The design and construction shall be such that the equipment is inherently reliable and can be easily maintained.

3.4.1 Reliability - The contractor shall implement a reliability program which (a) assures parts are applied within thermal, mechanical and electrical ratings and (b) provides necessary deratings in order to assure equipment reliability under the specified operating conditions.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:34Z
Check the source to verify that this is the current version before use.

3.4. 1.1 Operating Stability - The equipment shall operate as specified for not less than 1500 hours or 6 months, whichever comes first, without scheduled maintenance, realignment, or adjustment of any control not readily accessible to the operator during normal operation.

3.4.1.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity.

3.4.103 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment, including any built-in-test provisions, shall have a minimum acceptable (operating) MTBF (G1) of 750 hours when tested as specified in 4.4.3. The equipment shall have an inherent design reliability of 1,000 hours, (nominal) MTBF.

3.4.2 Maintainability - The design of the equipment shall include built-in-test features, modular construction and packaging, and provisions for test points .

3.4.3 Deleted.

3.4.4 interference Control - The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be control led within the limits of MIL-STD-461 unless otherwise specified.

3.4.5 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance .

3.4.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.7 Service Conditions - Service conditions (environmental) shall be in accordance with MIL-T-21200, C lass 2 except as specified herein.

3.4.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for marking laboratory bench tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

Temperature	Room ambient (30° 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input Power Voltage	108 to 118V AC, 400Hz 20Hz

3.5 Transportability - Transportability requirements shall be in accordance with MIL-T-2200.

3.6 Performance Characteristics -The performance of the equipment shall be as specified herein.

3.7 Detail Requirements

3.7.1 Test Set, Indicator Monitor - The test set shall meet the following requirements.

3.7.1.1 Function - The Indicator Monitor shall operate with the SM511IA/ASW to provide the capability of fully testing An/ASW-27B or equivalent Data Link terminals at the organizational level of maintenance. The indicator Monitor shall provide (1) the capability of transmitting preprogrammed (stored) control messages via the SM511A/ASW, (2) the capability of monitoring transmitted control or reply messages by displaying them on a 42 bit binary readout, and (3) the capability to display specific reply messages on appropriate numeric/discrete readouts.

3.7.1.2 Form Factor - The test shall be packaged in a combination case that shall meet the requirements of MIL-T-21200. The case shall have dimensions of 20.62 inches wide by 14.50 inches deep by 11.00 inches high excluding protrusions. The overall dimensions including protruding hardware shall be contained in an envelope 22.3 inches wide by 15.8 inches deep by 12.2 inches high.

3.7.1.3 Weight - The weight of the test set shall not exceed 53 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

3.7.1.4 Content -The test set shall contain the following major assemblies and subassemblies:

(1) Panel Assembly

(2) Display I Assembly (Circuit Card)

(3) Display II Assembly (Circuit Card)

(4) Display Ill Assembly (Circuit Card)

(5) Enclosure, RFI

(6) Case, Combination

(7) Power Supply

(8) Wiring Harness

(9) Discrete Assembly (Circuit Cord)

(10) Scale Conversion Assembly (Circuit Card)

(11) Multiplexer Assembly (Circuit Card)

(12) Control Assembly (Circuit Card)

(13) Message Generator Assembly (Circuit Card)

3.7.1.5 Controls - The test set shall have the following controls:

3.7.1.5.1 Power On/Off - A single toggle switch shall be provided to turn al I power to the Indicator Monitor test set on and off.

3.7.1.5.2 Monitor Mode - A seven position rotary switch shall be provided to select the following modes of operation:

(1) R-MSG - The reply message (R-MSG) mode is the primary mode of operation of the test set. In this mode, the indicator Monitor is operating on the RCVR Data input from an SM511A/ ASW. The test set will disply RO, R1, R3A, R3B and R3C reply messages on numeric and discrete readouts as defined in Table 3.7.1.10.2. With appropriate selections of Label Select switches and Reply Sub Label Select switches in this mode, 42 data bits of any reply message may also be displayed on the 42 bit binary readout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

(2) C-MSG -The control message (C-MS G) mode is a secondary mode of operation in which the test set displays 42 data bits of any control message on the 42 bit binary readout. The test set operates on the RCVR Data input from an SM511A/ASW in this mode and appropriate selection of A/C address thumbwheel switches and label select switches determines the control message to be displayed.

(3) Bench -1 - The Bench-1 mode, as the name implies, provides a capability of testing a Data Link terminal by direct cabling, i.e., bypassing the SM511A/ASW and the Data Link transmitter/receiver. In this mode the Radio Message Data Input signal (control message) is provided to the Data Link by a buffered signal from the Stored Message Generator. The test set then operates on the Radio Message Data Output signal (Reply Message) from the Data Link and processes reply messages as in the R-MSG mode.

(4) Bench -2 - Provision in the design shall be made to expand the Indicator Monitor capability to include the Bench-2 mode . In this mode the test set would receive a Demand Pulse from the AN/ASW-27B and clock out a serial control message from the Stored Message Generator. Upon receipt of a Frame Available Pulse from the AN/ASW-27B, the test set would clock in serial reply data, process, and display the reply data as in the R-MSG mode. The serial data transfer rates would be determined by the test set timing and would be approximate y 320 KHz.

(5) BIT- 1 - The BIT-1 mode shall provide sufficient built-in-test to indicate to the operator that the unit operates properly in the C-MSG mode with 8096 confidence. Up to five messages from the Stored Message Generator may be used to accomplish this built-in-test mode.

(6) BIT- 2 - The BIT-2 mode shall provide sufficient built-in-test to indicate to the operator that the unit operates properly in the R-MSG mode. The BIT-2 shall verify reply message conversion and display near zero scale of RO, R1, R3A, R3B, and R3C reply messages. Messages from the Stored Message Generator may be used to accomplish this built-in-test mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

(7) BIT-3 - The BIT-3 mode used in conjunction with BIT-2 shall provide sufficient built-in-test to indicate to the operator that the unit operates properly in the R-MSG mode with 80% confidence. The BIT-3 shall verify reply message conversion and display near full scale of RO, R1, R3A, R3B, and R3C reply messages. Messages from the Stored Message Generator may be used to accomplish this built-in-test mode. BIT-2 and BIT-3 modes combined may use up to eleven messages from the Stored Message Generator.

3.7.1.5.3 Reset - A momentary pushbutton switch shall be provided as a master reset for the test set, Whenever power is switched on for the test set, the reset switch shall be used to clear and initialize all logic in the test set, The reset switch is also the control that extinguishes all displays on the panel with the exception of XMTD MSG, R-MSG, and C-MSG, No Reply and Cancel Reply displays.

3.7.1.5.4 Stored Message Generator AUTO/OFF/MAN - A three position toggle switch shall provide for the following modes of operation for the Stored Message Generator.

(1) OFF - In this position there shall be no output from the Stored Message Generator though power may still be applied.

(2) MAN - The manual position shall cause the Stored Message Generator to output a message every 96 milliseconds on a continuous basis. The message transmitted is the stored message selected on the SELECTOR thumbwheels except when the Monitor Mode switch is in a BIT position. The transition of the AUTO/OFF/MAN switch from the OFF to MAN position shall initiate this mode.

(3) AUTO - The automatic position shall cause the Stored Message Generator to sequence through a series of messages determined by the Monitor Mode selection, the SELECTOR thumbwheel settings, and stored message in the read only memory of the message generator. In the following automatic modes the transition of the AUTO/OFF/MAN switch from the OFF to AUTO position shall initiate the sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

(a) Monitor Mode Switch not in a BIT Position - In these modes, the message generator shall output as the first message of a sequence the message selected on the SELECTOR thumbwheels. The first message will be transmitted every 96 milliseconds for a period of 7 +1 seconds. The message generator shall step to the next message in numeric order and transmit that message every 96 milliseconds for a period of 7 +1 seconds. The order continues until a recycle bit programmed into the memory is detected. Detection of the recycle bit shall cause the message generator to again output the message selected on the SELECTOR thumbwheels and after the 7 +1 second period step to the next message in numeric order. Should there be no recycle bits programmed, the message generator shall sequence in numeric order through message number 63 and then step to message number 00 and continue to step in numeric order from message number 00 through message number 63.

(b) Monitor Mode Switch in a BIT Position - In these modes, the message generator shall start with the first message of the BIT sequence selected on the Monitor Mode switch and cycle through the last message of the BIT sequence selected then stop. The BIT sequence may only be repeated by placing the AUTO/OFF/MAN switch in OFF and then returning to the AUTO position.

3.7.1.5.5 Stored Message Generator SELECTOR - Two thumbwheel switches shall provide for selection of stored messages in MAN and AUTO modes of the Stored Message Generator as detailed in paragraphs 3.7.1.5.4 (2) and 3.7.1.5.4 (3), respectively.

3.7.1.5.6 Aircraft Address - A group of five octal -coded thumbwheel switches (0,0,0,0,0 to 1,7,7,7,7) shall be provided to select the Aircraft (A/C) Address for control messages. The A/C address switches serve a dual function in the test set operation. These switches determine the aircraft address code inserted into each control message generated by the Stored Message Generator. The second function is to provide for processing only those control messages which have the same aircraft address code as selected by the thumbwheel switches.

3.7.1.5.7 Label Select - A set of five toggle switches shall be provided for selecting the label of message data to be displayed on the 42 bit binary readout. The five switches shall be binary weighted 2^0 (1) through 2^4 (16). All five switches shall apply when selecting the display of control messages and on} y the 2^0 through 2^3 switches shall apply when selecting the display of reply messages.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:34Z
Check the source to verify that this is the current version before use.

3.7.1.5.8 Reply Sub Label Select -A five position rotary switch shall be provided for selecting an A, B,C, or D sub label of reply message data to redisplayed on the 42 bit binary readout. In the OFF position, reply message data to be displayed will be determined only by the Label Select switches.

3.7.1.5.9 Display Update/Hold - A toggle switch shall be provided to al low holding of the last received data in storage and on displays. Any message in process at the time this switch is placed into the Hold position will be completely processed and held in storage and on the displays. While in the Hold position, activating the Reset switch of paragraph 3.7.1.5.3 will clear, i.e., extinguish, the data played. The Update position of this switch will al low displays to be updated by received message data. When this switch is changed from the Hold to the Update position, the Reset must be activated to enable the test set to start the continual update process.

3.7.1.5.10 Clear - A momentary pushbutton switch shall be provided to clear the no Reply and Cancel Reply status indicators.

3.7.1.6 Electrical Connections - Connections to external circuits shall be as follows:

Reference Designation	Receptacle Type	Function
J1	M39012/16-007	Receiver Data
J2	M23329/4-17	Stored Message Data
J3	MS3124E-16-26P	Data Link Data
J4	M23329/4-17	Record Data
J5	MS3124E-14-4P	Power

3.7.1.7 Input Signals - The Indicator Monitor shall operate on input signals as follows:

3.7.1.7.1 Receiver Data - This signal is the VHF receiver output of the SM511A/ ASW. The NRZ Control and Reply Message data as defined by the Message Standard WR-28A will be provided by this signal, The receiver shall provide a 550 +150 millivolt peak-to-peak signal for deviation of +20 kHz when loaded by a 1000 ohm load (shunted by 100 picofarad maximum). The output shall be positive for a +20 kHz deviation at the receiver input. The signal shall be low-pass filtered. A DC restoration and zero crossover detector shall also be provided to al low bit detection under low signal-to-noise ratio conditions.

3.7.1.7.2 Radio Message Data Output - This signal is the NRZ Reply Message data output of the AN\ASW-27B. The Reply Message data contained in this signal is defined by WR-28A. A binary ONE is indicated by +5.0 +1.0 V DC and a binary ZERO by -5.0 +1.0 V DC.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:34Z
Check the source to verify that this is the current version before use.

3.7.1.8 Output Signals - The indicator Monitor shall provide output signals as follows.

3.7.1.8.1 Stored Message Data - Control Messages composed of synchronization pattern, aircraft address, data fields, and parity bits as specified in WR-28A shall be provided as an output for use by the SM511A/ASW and AN/ASW-27B. This signal is the buffered output of the Stored Message Generator and the occurrence is determined by the particular operating mode as detailed in Paragraph 3.7.1.5.4. A binary ONE shall be represented by +5.0 +1.0 V DC and a binary ZERO by -5.0 +1.0 V DC .

3.7.1.8.2 Record Data - This signal shall be the Receiver Data input signal conditioned and buffered for use by an external recorder, A binary ONE shall be represented by +5.0 +1.0 V DC and a binary ZERO by -5.0 +1.0 V DC.

3.7.1.9 Message Synchronization - The synchronization of messages shall consist of (1) recognition of the unique synchronization pattern in a Control Message as specified in WR-28A, (2) correlation of the Control Message aircraft address code with the code selected on the A/C thumbwheels, (3) detection of the correct Control Message direction bit defined in WR-28A, (4) recognition of the unique synchronization pattern in the Reply Message 30 to 40.4 milliseconds after the end of the properly addressed Control Message, and (5) detection of the correct Reply Message direction bit defined in WR-28A. The synchronization and subsequent processing of a Reply Message shall be completely dependent upon prior receipt of an appropriate Control Message.

3.7.1.10 Message Processing - The processing of messages in the test set shall consist of (1) storage and display of 42 data bits for either Control or Reply Messages, (2) detection, storage and display of Control Message parity, (3) storage and display of Control Message cancel reply bit, and (4) appropriate conversion, storage and display of RO, R1, R3A, R3B, and R3C Reply Messages.

3.7.1.10.1 C-MSG Processing - In this mode of operation, the 42 data bits of Control Messages with aircraft address and label codes corresponding to selected codes on A/C address thumbwheels and label select switches respectively and with correct direction bit will be transferred to a holding register and displayed on the 42 bit binary readout. The test set shall check the three parity segments of Control Messages and light the CMD error indicator if one or more of the parities is incorrect. The 42 data bits shall be displayed irrespective of parity errors.

3.7.1.10.2 R-MSG Processing - in this mode of operation, the test set shall first synchronize on a properly addressed Control Message. The cancel reply bit of the Control Message shall be stored and an appropriate indication given on the Cancel Reply indicator. The test set shall time the period, 30-40.4 milliseconds, in which a Reply Message should occur, irrespective of the Control Message cancel reply bit. If no Reply Message occurs during this window or if the Reply Message direction bit is incorrect, the No Reply indicator shall be lit. The test shall not process a Reply Message with an incorrect direction bit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

For any Reply Message which has a label corresponding to the code on Label Select switches and sublable corresponding to the code selected on Reply Sub Label Select switch (if any), the test set shall transfer the 42 data bits to a holding register and displayed on the 42 bit binary readout.

RO, R1, R3A, R3B, and R3C messages shall reconverted from binary form to Numeric (BCD) or discrete coding and displayed on readouts. Table 3.7.1.10.2 lists the conversion requirements for each of these five Reply Message types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

Table 3.7.1.10.2(c)
Reply Message Conversion and Display Requirements

Information	No. of Bits	Conversion	Display Type, Range and Resolution (See Note 3)
Replay Message R_0			
A/C Heading	8	Binary to BCD	Numeric, 0--359 degrees, 1 degree
A/C Altitude	8	Binary to BCD	Numeric, 0-12,700 feet, 100 feet and 0-127,000 feet, 1000 feet
A/C Airspeed	8	Binary to BCD	Numeric, 0-3585 knots, 14 knots
Fuel Status	6	Binary to BCD	Numeric, 0-25,200 lbs, 400 lbs
A/C Type	3	None	Numeric, 0-7, 1
Weapons Status	4	None	Discrete (two groups of two), 0-15 binary
Reply Message R_1			
TACAN Station	7	Binary to BCD	Numeric, 0-126, 1
Y Mode	1	None	Discrete, 0-1 binary
TACAN Bearing	9	Binary to BCD	Numeric, 0-359 degrees, 1 degree
TACAN Range	9	Binary to BCD	Numeric, 0-306, 6 naut. miles, .6 naut. mile
Weapons Status	4	None	Discrete (two groups of two) 0-15 binary
Pilot Ready	1	None	Discrete, 0-1 binary
Autopilot Engaged	1	None	Discrete, 0-1 binary
Discretes	5	Binary to BCD	Numeric, 0-31,1
Reply Message R3A			
Target X Position	8	Mod. 2's Comp, Binary to BC D (See Note 1)	Numeric, -128 to +127 naut. miles, 1 naut. mile
Target Y Position	8	Mod. 2's Comp, Binary to BC D (See Note 1)	Numeric, -128 to +127 naut. miles, 1 naut. mile
Target Altitude	7	Binary to BCD	Numeric, 0-127,000 feet 1,000 feet
Target ID	2	None	Numeric, 0-4,1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

Information	No. of Bits	Conversion	Display Type, Range, and Resolution (See Note 3)
Target ID	1	None	Discrete, 0-1 binary
Discretes	3	None	Numeric, 0-7, 1
Discretes Parity	1	None	Discrete, 0-1 Binary
Target Report	1	None	Discrete, 01 binary
Not TWS	1	None	Discrete, 0-1 binary
Pointer	1	None	Discrete, 0-1 binary
Raid Size	2	None	Numeric, 0-3,1
Reply Message R3B			
A/C Latitude	16	Mod .2's Comp. Binary to BCD (See Note 1)	Numeric/Discrete (See Note 2), N89° 59.9' to S90° 00.0', .1'
A/C Longitude	17	Mod .2's Comp. Binary to BCD (See Note 1)	Numeric/Discrete (See Note 2), E179° 59.9' to W180° 00.0', .1'
Reply Message R3C			
Target X Velocity	9	Mod .2's Comp. Binary to BCD (See Note 1)	Numeric, -4096 to +4080 knots, 16 knots
Target Y Velocity	9	Mod .2's Comp. Binary to BCD (See Note 1)	Numeric, -4096 to +4080 knots, 16 knots

Note 1- Modified 2's complement binary is defined as sign bit = 1 - straight binary and sign bit = 0-2's complement.

Note 2 - N, S, E, and Y may be indicated by a discrete indicator.

Note 3 -The following are acceptable conversion tolerances:

A/C Heading	± 1 deg
A/C Altitude	± 1000 ft or + 100 ft as applicable
A/C Airspeed	± 1 knot
Fuel Status	± 100 lb
TACAN Bearing	± 1 deg
TACAN Range	± 0.1 N.M.
TGT X Pos.	± 1 N.M.
TGT Y peso	± 1 N.M.
TGT Altitude	± 1000 ft.
Latitude	± 0.2 min
Longitude	± 0.2 min
TGT X Vel	± 1 knot
TGT Y Vel	± 1 knot

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:34Z
Check the source to verify that this is the current version before use.

3.7.1.11 Indicators - In addition to the indicators defined in paragraph 3.7.1.10.1, paragraph 3.7. 1.10.2, and Table 3.7.1.10.2, the indicator Monitor shall have the following displays.

(1) RO, R1, R3A, R3B, and R3C- Discrete displays that indicate the last reply message received in R-Msg mode was a RO, R1, R3A, R3B, or R3C message.

(2) R-Msg and C-Msg - Discrete displays that indicate the data bits displayed on the 42 bit binary readout are Reply or Control Message data bits.

(3) XMTD MSG -Numeric displays that indicate the message number being read out of the Stored Message Generator Memory.

(4) POWER -Discrete display that indicates primary power is applied and the power switch is in the ON position.

3.7.1.12 Electrical Power Source -The test set shall be capable of satisfactory operation when supplied input power having characteristics within the Normal Steady-State Limits (NSSL) specified for Category B equipment in MIL-STD-704. The test set shall not be damaged and shall return to normal operation after being subjected to voltage transients and Abnormal Steady-State Limits (ASSL) also specified for Category B equipment in MIL-STD-704.

3.8 Dimensions - The maximum overall size shall not exceed 22.3 inches wide by 12.2 inches high by 15.8 inches long.

3,9 Finish - The exterior surfaces of the combination case shall be primed in accordance with MIL-T-21200. The final cost of paint shall be gray wrinkle enamel per MIL-E-5558, Type 1, Color Number 512 per ANA Bulletin 166.

3.10 Interchangeability - interchangeability shall be in accordance with MIL-T-21200.

3.11 Workmanship - Workmanship shall be in accordance with MIL-T-21200.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

First Article (preproduction) tests

Initial production tests

Acceptance tests

Life test

4.2 First Article (Preproduction) Tests - First article (reproduction) tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain i-he services of a commercial testing laboratory acceptable to the Government.

4.2.1 First Article (Reproduction) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental and interference tests. Environmental tests shall consist only of Temperature-Altitude, Vibration, Humidity, Salt Fog, and Shock tests and shall be conducted in accordance with the requirements of Specification MIL-T-5422 (Except humidity will be limited to 50°C maximum temperature) and limits specified in Specification MIL-T-21200. Interference tests shall be conducted in accordance with MIL-STD-462 to the requirements of MIL-STD-461.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:34Z
Check the source to verify that this is the current version before use.

4.2.3 First Article (Preproduction) Approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the test sample. Pre-fabrication of production equipment prior to the approval of the test sample is at the contractor's own risk.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved test sample. Each equipment shall be capable of successfully passing the same tests as imposed on the test sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipment shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successful y passed all individual tests. The test sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish al I samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual tests (4.4.1)

(2) Sampling Tests (4.4.2)

(3) Reliability Assurance Tests (4.4.3)

(4) Special Tests (4.4.4)

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product (4.4.1.1)

(2) Operational Test (4.4.1.2)

(3) Manufacturing Run in Test (4.4.1.3)

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

NOTE: Approval of the operational test procedure must be obtained from the Naval Air Systems Command prior to testing.

4.4.1.3 Manufacturing Run in Test -Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency (within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's,

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operating within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to Be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

* NOTE: If by contract action the initial production test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, marking all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions.. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Not Applicable.

4.4.4 Special Tests. - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in 4.5.1. The life test shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient
Altitude	Normal ground (0-5000 ft)
Humidity	Room Ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	27.0 ±2.0 volts
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure .

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:34Z
Check the source to verify that this is the current version before use.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5. Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned os follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all “wear” items. The “wear” items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

5.2 Marking - in addition to any special marking by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use -The Indicator Monitor Test Set shall be used to provide organizational maintenance support to Link 4A Airborne Digital Data Communications Equipment with Reply Message capability. The test set will enable the operator to transmit Control Messages to the Data Link equipment and receive Reply Messages via the SM511A/ ASW. The test set will then display the Reply Message for ease of verifying proper Data Link operation or troubleshooting malfunctions.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanies by complete supporting information.

6.3 Nonrepairable Subassemblies - As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly y should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied), the Non-repair-able subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:34Z
Check the source to verify that this is the current version before use.

(2) This Specification -This specification shall have precedence over all applicable subsidiary specification. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1).

6.6 Type Designation - The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Digital Data Simulator (SM511A/ASW)

(2) Digital Data Recorder-Reproducer (RD-330/ASM)

6.9 Specification Cognizance - This specification is under the cognizance of AIR-53421B (Engineers Code).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

6.10 Definitions - The following table for an eight bit coded value illustrates straight binary and modified two's complement coding.

	Straight Binary		Mod 2 's Comp.	
Value	Sign	Value	Sign	Value
+127	1	1111111	1	1111111
- - -	-	- -	-	- -
+1	1	0000001	1	0000001
0	1	0000000	1	0000000
-1	0	0000001	0	1111111
"2	0	0000010	0	1111110
		- - -		- - -
-127	0	1111111	0	0000001
-128			*0	0000000

*This code does not occur in straight binary coding. In Modified Two's Complement this code takes on a value of one increment greater than most negative binary code.

Custodians:

Navy-AS

Preparing Activity:

Navy-AS

Project No. 4935-N126

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-I-81987(AS) INDICATOR MONITOR TEST SET ID 1956/ASW

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U S GOVERNMENT PRINTING OFFICE 1973—714-866/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:34Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20361

POSTAGE AND FEES PAID
DOD 316

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NAVAL AIR SYSTEMS COMMAND(AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20361

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:34Z
Check the source to verify that this is the current version before use.